CHAPTER 5

# Sliding Window Protocols

Stop-and-wait · Go-Back-N · Selective Repeat · efficiency · window size · sequence-number bits

**QUICK TRICK**

Let **a = $T_p$ / $T_t$**. Stop-and-wait efficiency **η = 1 / (1 + 2a)**. Window W: **η = min(1, W / (1 + 2a))**. To fill the pipe use **W = 1 + 2a**.
Sequence bits k: **GBN** sender window ≤ $2^k - 1$. **SR** window ≤ $2^{k-1}$.

## 5.1 Why a window?

● **LEVEL 1 · BEGINNER**

**The problem.** Reliable delivery needs acknowledgements (ACKs). Waiting for an ACK after every frame wastes the link when the delay is long. **The idea.** Let the sender keep up to **W frames in flight** before it must wait.

**Analogy.** A conveyor belt of parcels. Stop-and-wait puts one parcel on the belt and waits for the receipt. Sliding window keeps the belt full.

## 5.2 Stop-and-wait

● **LEVEL 2 · EXAM**

```
time ->   |-- Tt --|---------- Tp ---------->|            (frame travels to receiver)
                                             |<--------- Tp ----------|   (ACK travels back)
Sender sends frame 1 ........ idle (waiting) ........ ACK arrives -> sends frame 2
Cycle time = Tt + Tp + Tp = Tt + 2*Tp          (ACK transmission time neglected)
Efficiency = Tt / (Tt + 2Tp) = 1 / (1 + 2a)    Throughput = efficiency x bandwidth
```

- If ACK transmission time $T_{ack}$ is given: cycle = $T_t + 2T_p + T_{ack}$.
- With frame loss probability p: η = (1 − p) / (1 + 2a).

## 5.3 Sliding window

● **LEVEL 2 · EXAM**

**Frames per RTT:** the sender can send W frames in each cycle of (1 + 2a) frame-times, so **η = W / (1 + 2a)** and it cannot exceed 1. **Smallest W that gives 100%** is **1 + 2a** (round *up* to an integer).

| | Stop-and-wait | Go-Back-N (GBN) | Selective Repeat (SR) |
|---|---|---|---|
| Sender window | 1 | ≤ $2^k - 1$ | ≤ $2^{k-1}$ |
| Receiver window | 1 | 1 | = sender window ($2^{k-1}$) |
| ACK type | each frame | Cumulative | Individual |
| Out-of-order frame | n/a | Discarded | Buffered |
| On loss / timeout | resend 1 frame | Resend the lost frame **and all after it** | Resend **only** the lost frame |
| Receiver buffer | none | none | needed |
| Min sequence bits for window W | 1 | k = ⌈$\log_2(W + 1)$⌉ | k = ⌈$\log_2(2W)$⌉ |
| Efficiency (error p, W ≥ 1+2a) | (1−p)/(1+2a) | (1−p)/(1+2ap) | 1 − p |

```
GBN, window 7, frame 3 lost
Sender  : 0 1 2 3 4 5 6  ...(timer for 3 expires)...  3 4 5 6 (resent)
Receiver: ack0 ack1 ack2   4,5,6 arrive out of order -> DISCARDED (ack2 repeated)
SR      : same loss, but receiver BUFFERS 4,5,6 and only 3 is resent.
```

**QUICK TRICK**

Why SR needs $W \le 2^{k-1}$: the old window and new window must never overlap, or a repeated old frame looks like a new one. Why GBN allows $2^k - 1$: the receiver window is only 1, so one number is kept spare.

## 5.4 Worked example

● **LEVEL 2 · EXAM**

Link 1 Mbps, frames 1000 bits, one-way propagation 10 ms. $T_t$ = 1 ms, $T_p$ = 10 ms, so **a = 10**.

- Stop-and-wait: η = 1/21 = **4.76%**; throughput = 47.6 kbps.
- Window W = 8: η = 8/21 = **38.1%**.
- Full use needs W = 1 + 2a = **21**.
- GBN bits: $2^k - 1 \ge 21 \rightarrow k =$ **5** (31). SR bits: $2^{k-1} \ge 21 \rightarrow k - 1 = 5 \rightarrow k =$ **6**.

## 5.5 Common mistakes

**COMMON MISTAKE**

- Using window = $2^k$ for GBN (it is $2^k - 1$).
- Using propagation delay one-way for the cycle. The cycle uses $T_t + \mathbf{2}T_p$.
- Not rounding W up. If 1 + 2a = 10.4, you need 11 frames.
- Confusing the *window size* with the *number of sequence numbers*.
- Forgetting that efficiency above 100% is impossible: cap at 1.

## GATE QUICK CHECK — Chapter 5

```
a = Tp/Tt       SW eff = 1/(1+2a)       Window eff = min(1, W/(1+2a))
W_opt = 1 + 2a (ceil)     GBN: W <= 2^k - 1     SR: W <= 2^(k-1)
GBN bits k = ceil(log2(W+1))     SR bits k = ceil(log2(2W))
With ack time: cycle = Tt + 2Tp + Tack          Throughput = eff x link rate
```

### Practice questions

● **LEVEL 1 · BEGINNER**

**Q1.** Stop-and-wait with $T_t$ = 1 ms and $T_p$ = 4 ms. Efficiency?

**Q2.** Maximum sender window for GBN with 3 sequence bits.

**Q3.** Maximum window for SR with 4 sequence bits.

**Q4.** Which protocol buffers out-of-order frames at the receiver?

● **LEVEL 2 · EXAM**

**Q5.** Optimal window if a = 15, and minimum sequence bits for GBN and SR.

**Q6.** 2 Mbps link, 1000 B frames, one-way propagation 20 ms. Stop-and-wait efficiency and throughput.

**Q7.** Window needed to reach 100% in the previous question, and sequence bits for GBN and SR.

**Q8.** Stop-and-wait with a = 4 and frame loss probability 0.2. Efficiency?

● **LEVEL 3 · GATE**

**Q9.** GBN, window 7, frames 0 to 6 sent, frames 0–2 acknowledged, frame 3 lost. How many frames are retransmitted after the timeout?

**Q10.** A satellite link: 1 Mbps, 1000-bit frames, RTT 500 ms. Window and sequence bits for GBN and SR at 100% use?

**Q11.** Why can SR not use window $2^k$?

**Q12.** Window W = 6 on a link where 1 + 2a = 21. Efficiency?

## Answers

| Q | Solution |
|---|---|
| 1 | $1/(1 + 2\times4) = 1/9 \approx$ **11.1%**. |
| 2 | $2^3 - 1 =$ **7**. |
| 3 | $2^3 =$ **8**. |
| 4 | **Selective Repeat**. |
| 5 | $W = 1 + 30 =$ **31**. GBN: $2^k - 1 \geq 31 \rightarrow$ **k = 5**. SR: $2^{k-1} \geq 31 \rightarrow k - 1 = 5 \rightarrow$ **k = 6**. |
| 6 | $T_t = 8000/2\times10^6 = 4$ ms; cycle = 4 + 40 = 44 ms. $\eta = 4/44 =$ **9.09%**; throughput = 0.0909 × 2 Mbps ≈ **181.8 kbps**. |
| 7 | $a = 20/4 = 5$, so $W = 1 + 10 =$ **11**. GBN: $2^k - 1 \geq 11 \rightarrow$ **k = 4**. SR: $2^{k-1} \geq 11 \rightarrow$ **k = 5**. |
| 8 | $(1 - 0.2)/(1 + 8) = 0.8/9 \approx$ **8.89%**. |
| 9 | Frames 3, 4, 5, 6: **4 frames**. |
| 10 | $T_t = 1$ ms, $2T_p = 500$ ms, $W =$ **501**. GBN: $2^k - 1 \geq 501 \rightarrow$ **k = 9**. SR: $2^{k-1} \geq 501 \rightarrow k - 1 = 9 \rightarrow$ **k = 10**. |
| 11 | Windows of old and new frames would overlap, so a retransmitted old frame could be accepted as new data. |
| 12 | 6/21 = **28.6%**. |

**Next:** Chapter 6 — Medium Access Control (ALOHA, CSMA, CSMA/CD).

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